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3.2.1 | Strut to Tree Connection

Use Simpson Strong Tie online selection tool.





Object Visibility

☒ Carrying

☒ Carried

☒ Hanger

OUTPUT

JOB LIST (1)

Add to Job list

DOWNLOAD CSV

Search Table

Q

Show Optimized Models ☐

Showing 1 entry

	Model	Installed Cost	Width	Height	Bearing	TF Depth	TF Fasteners	Face Fasteners	Joist Fasteners	Download (lbs)	Uplift (lbs)
	HUC46TFX SLU45	Lowest	3.563	5.375	2.5	2.5	(4) 0.162" x 3 1/2"	(6) 0.162" x 3 1/2"	(4) 0.148" x 3"	1845	575

Table Notes

1. All loads are displayed in units of pounds and based on Allowable Stress Design

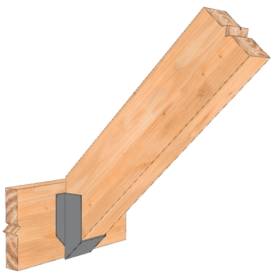
2. Click on the Models above to be taken to the product page for more information, refer to the current Wood Construction Connectors catalog for [General Notes](#) and [Installation Instructions](#)

Fig. 1 - Option 1 | time: no time



HS

3D VIEWER



Object Visibility

☒ Carrying

☒ Carried

☒ Hanger

OUTPUT

JOB LIST

Add to Job list

DOWNLOAD CSV

Search Table

Q

Show Optimized Models ☐

Showing 10 entries

	Model	Installed Cost	Width	Height	Bearing	TF Depth	TF Fasteners	Face Fasteners	Joist Fasteners	Download (lbs)	Uplift (lbs)
☐	 U46X SLU45	Lowest	3.563	4.875	2	-	-	(8) 0.148" x 3"	(4) 0.148" x 3"	1000	400
☒	 U46X SLU45	Lowest	3.563	4.875	2	-	-	(8) 0.162" x 3 1/2"	(4) 0.148" x 3"	1090	400

Fig. 2 - Option 2 | time: no time



3.2.2 | Top rail Corner

Use AWC online connection tool.

←

ConnectionCalc Results

Analysis Type

Design Method:

?

Allowable Stress

Connection Loading:

?

Lateral

Fastener Type:

?

Wood Screw

Main Member Parameters

Main Member material category:

?

Solid Lumber/Timber

Type:

?

Douglas Fir-Larch

Main Member Thickness:

?

1-1/2 in

Load to Grain Angle:

?

0 deg

Side Member Parameters

Side Member material category:

?

Steel

Side Member Thickness:

?

14 gage

Wood Screw Parameters

Size:

?

No.8

Length:

?

1-1/2 in

Analysis Factors

Load Duration (CD):

?

1

Wet Service (CM):

?

0.7

End Grain (Ceg):

?

1

Temperature Factor (Ct):

?

0.8

Results

Adjusted ASD Capacity:

?

55 lb.

Yield Modes

Fig. 3 - Top Rail - Corner Plate Input



← ConnectionCalc Results	
②	1
Temperature Factor (C):	
②	0.8
Results	
Adjusted ASD Capacity:	55 lb.
Yield Modes	
Z' Im:	
②	195 lb.
Z' Is:	
②	156 lb.
Z' Il:	
②	78 lb.
Z' Illm:	
②	89 lb.
Z' Ills:	
②	55 lb.
Z' IV:	
②	72 lb.
Notes	
Douglas Fir- Larch Main Member Dowel Bearing Strength (psi):	
②	4,650 psi
Side Member Assumed Steel Alloy:	
②	ASTM A653 G33
Steel Side Member Dowel Bearing Strength (psi):	
②	61,850 psi
Fastener Bending Yield Strength:	
②	90,000 psi
Tip length:	
②	2x diameter
Screw Dowel Bending Strength rounding:	
②	nearest 50 psi
Lateral Results Rounding:	
②	nearest 10 lb
Toe-nails in wood members:	
②	results do not apply
Disclaimer:	
While every effort has been made to insure the accuracy of the information presented, and special effort has been made to assure that the information reflects the state-of-the-art, neither the American Wood Council nor its members assume any responsibility for any particular design prepared from this Connection Calculator. Those using this Connection Calculator assume all liability from its use.	

Fig. 4 - Top Rail - Corner Plate Capacity

Use 4-#8 screws = 55 lbs * 4 = Capacity 220 lbs | Demand = 200 lbs.

